Normal Distribution & Probability

Statistics Worksheet · Grades 11–12

Name: _____ Date: _____

Learning Objectives

- Convert raw scores to z-scores using the standardization formula
- Find probabilities using the standard normal distribution (normalcdf)
- Interpret left-tail, right-tail, and between-values probabilities in context

Problems

1. The average height of students at Lincoln High is 67 inches with a standard deviation of 2.5 inches. Find the z-score for a student who is 72 inches tall.

$$z = \frac{x - \mu}{\sigma} = \frac{72 - 67}{2.5}$$

2. Using the same distribution (mean = 67, standard deviation = 2.5), find the z-score for a student who is 60 inches tall.

$$z = \frac{60 - 67}{2.5}$$

3. A normally distributed data set has a mean of 50 and a standard deviation of 8. Match each raw score to its correct z-score by completing the table below.

Raw Score (x)	z-score
34	
50	
58	
66	

4. The heights of students at Barstow High are normally distributed with a mean of 67 inches and a standard deviation of 2.5 inches. What is the probability that a randomly selected student is shorter than 60 inches? Use the normalcdf function with the standardized z-score.

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$$P(X < 60) = P\left(Z < \frac{60 - 67}{2.5}\right) = P(Z < -2.8)$$

5. Explain why the probability that a student is EXACTLY 73 inches tall — in a continuous normal distribution — is equal to zero. Then find the z-score for 73 inches using mean = 67 and standard deviation = 2.5.

$$z = \frac{73 - 67}{2.5}$$

6. Using mean = 67 inches and standard deviation = 2.5 inches, find the probability that a student is taller than 71 inches. Shade the correct region and write the normalcdf syntax you would enter in a TI-84.

$$P(X > 71) = P\left(Z > \frac{71 - 67}{2.5}\right) = P(Z > 1.6)$$

7. The heights at Barstow High follow a normal distribution with mean 67 inches and standard deviation 2.5 inches. Find the probability that a randomly selected student has a height between 61 inches and 71 inches.

$$P(61 < X < 71) = P\left(\frac{61 - 67}{2.5} < Z < \frac{71 - 67}{2.5}\right)$$

8. A student claims that about half the students at Barstow High are shorter than 67 inches. Using the properties of a normal distribution with mean 67 and standard deviation 2.5, confirm or refute this claim by calculating P(X less than 67).

$$P(X < 67) = P\left(Z < \frac{67 - 67}{2.5}\right) = P(Z < 0)$$

9. Test scores at a school are normally distributed with a mean of 78 and a standard deviation of 6. Find the probability that a randomly selected student scored between 70 and 88. Also identify how many standard deviations each boundary is from the mean.

$$P(70 < X < 88) = P\left(\frac{70 - 78}{6} < Z < \frac{88 - 78}{6}\right)$$

10. Weights of adult males in a city are normally distributed with a mean of 180 pounds and a standard deviation of 15 pounds. A health clinic defines 'underweight' as below 150 pounds and 'overweight' as

above 210 pounds. Find (a) the probability a randomly selected male is underweight, (b) the probability he is overweight, and (c) the probability he falls in the healthy range between 150 and 210 pounds.

$$P(X < 150), \quad P(X > 210), \quad P(150 < X < 210)$$

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Normal Distribution & Probability — Answer Key

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Answer Key

1. Answer: z = 2.0

- Identify the values: x = 72, μ = 67, σ = 2.5
- Substitute into the z-score formula: z = (72 – 67) / 2.5
- Simplify: z = 5 / 2.5 = 2.0

2. Answer: z = -2.8

- Identify the values: x = 60, μ = 67, σ = 2.5
- Substitute: z = (60 – 67) / 2.5 = –7 / 2.5
- Simplify: z = –2.8

3. Answer: See completed table

Raw Score (x)	z-score
34	-2.0
50	0.0
58	1.0
66	2.0

- Use z = (x – μ) / σ for each value with μ = 50 and σ = 8
- z = (34 – 50) / 8 = –16 / 8 = –2.0
- z = (50 – 50) / 8 = 0 / 8 = 0.0
- z = (58 – 50) / 8 = 8 / 8 = 1.0
- z = (66 – 50) / 8 = 16 / 8 = 2.0

4. Answer: P(X < 60) ≈ 0.0026 or about 0.26%

- Compute z = (60 – 67) / 2.5 = –2.8
- Use normalcdf(–1E99, –2.8, 0, 1) on the TI-84
- The calculator returns approximately 0.0026
- Interpret: about 0.26% of students are shorter than 60 inches

5. Answer: P(X = 73) = 0; z = 2.4

- In a continuous distribution, the probability of any single exact value is 0 because a single point has no area under the curve
- Compute z = (73 – 67) / 2.5 = 6 / 2.5 = 2.4
- Although z = 2.4, P(X = 73) = 0 for a continuous distribution

6. Answer: P(X > 71) ≈ 0.0548 or about 5.48%

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- Since we want P(Z > 1.6), the shaded region is to the right
- TI-84 syntax: normalcdf(1.6, 1E99, 0, 1)
- Result ≈ 0.0548, so about 5.48% of students are taller than 71 inches

7. Answer: P(61 < X < 71) ≈ 0.9104 or about 91.04%

- Compute lower z: $z_1 = (61 – 67) / 2.5 = –2.4$
- Compute upper z: $z_2 = (71 – 67) / 2.5 = 1.6$
- TI-84 syntax: normalcdf(–2.4, 1.6, 0, 1)
- Result ≈ 0.9104, meaning about 91.04% of students have heights in this range

8. Answer: P(X < 67) = 0.5000 — the claim is correct

- Compute z = (67 – 67) / 2.5 = 0
- In a normal distribution, P(Z < 0) = 0.5000 exactly
- The mean is the center of the distribution, so exactly 50% of values fall below it
- The student's claim is correct

9. Answer: P(70 < X < 88) ≈ 0.8164 or about 81.64%

- Compute lower z: $z_1 = (70 – 78) / 6 = –8 / 6 ≈ –1.33$ (about 1.33 standard deviations below the mean)
- Compute upper z: $z_2 = (88 – 78) / 6 = 10 / 6 ≈ 1.67$ (about 1.67 standard deviations above the mean)
- TI-84 syntax: normalcdf(–1.33, 1.67, 0, 1)
- Result ≈ 0.8164, so about 81.64% of students scored between 70 and 88

10. Answer: (a) ≈ 0.0228 or 2.28%; (b) ≈ 0.0228 or 2.28%; (c) ≈ 0.9545 or 95.45%

- Compute z for 150: z = (150 – 180) / 15 = –30 / 15 = –2.0
- Compute z for 210: z = (210 – 180) / 15 = 30 / 15 = 2.0
- (a) P(X < 150) = normalcdf(–1E99, –2.0, 0, 1) ≈ 0.0228
- (b) P(X > 210) = normalcdf(2.0, 1E99, 0, 1) ≈ 0.0228
- (c) P(150 < X < 210) = normalcdf(–2.0, 2.0, 0, 1) ≈ 0.9545
- Note: This confirms the empirical rule — approximately 95.45% of data falls within 2 standard deviations of the mean

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